

Practice questions Binomial distribution

1. A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.
 - i. The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
 - ii. Find mean (expected value) and variance
2. The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly 5 of the next 7 patients having this operation survive?
3. A nationwide survey of college seniors by the University of Michigan revealed that almost 70% disapprove of daily pot smoking, according to a report in Parade. If 12 seniors are selected at random and asked their opinion, find the probability that the number who disapprove of smoking pot daily is
 - (a) Anywhere from 7 to 9;
 - (b) At most 5;
 - (c) Not less than 8.
4. In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that
 - (a) From 3 to 6 have blowouts;
 - (b) Fewer than 4 have blowouts;
 - (c) More than 5 have blowouts.
5. A particularly long traffic light on your morning commute is green 20% of the time that you approach it. Assume that each morning represents an independent trial.
 - (a) Over five mornings, what is the probability that the light is green on exactly one day?
 - (b) Over 20 mornings, what is the probability that the light is green on exactly four days?
 - (c) Over 20 mornings, what is the probability that the light is green on more than four days?
6. Because not all airline passengers show up for their reserved seat, an airline sells 125 tickets for a flight that holds only 120 passengers. The probability that a passenger does not show up is 0.10, and the passengers behave independently.
 - (a) What is the probability that every passenger who shows up can take the flight?
 - (b) What is the probability that the flight departs with empty seats?

Practice questions Poisson distribution

1. Contamination is a problem in the manufacture of optical storage disks. The number of particles of contamination that occur on an optical disk has a Poisson distribution, and the average number of particles per centimeter squared of media surface is 0.1. The area of a disk under study is 100 squared centimeters. Find the probability that 12 particles occur in the area of a disk under study.
2. The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour.
 - a. What is the probability that there are exactly 5 calls in one hour?
 - b. What is the probability that there are 3 or less calls in one Hour?
 - c. What is the probability that there are exactly 15 calls in two hours?
 - d. What is the probability that there are exactly 5 calls in 30 minutes?
3. The number of flaws in bolts of cloth in textile manufacturing is assumed to be Poisson distributed with a mean of 0.1 flaws per square meter.
 - i. What is the probability that there are two flaws in 1 square meter of cloth?
 - ii. What is the probability that there is one flaw in 10 square meters of cloth?
 - iii. What is the probability that there are no flaws in 20 square meters of cloth?
 - iv. What is the probability that there are at least two flaws in 10 square meters of cloth?
4. The number of surface flaws in plastic panels used in the interior of automobiles has a Poisson distribution with a mean of 0.05 flaws per square foot of plastic panel. Assume an automobile interior contains 10 square feet of plastic panel.
 - a. What is the probability that there are no surface flaws in an auto's interior?
 - b. If 10 cars are sold to a rental company, what is the probability that none of the 10 cars has any surface flaws?
 - c. If 10 cars are sold to a rental company, what is the probability that at most one car has any surface flaws?
5. The probability that your call to a service line is answered in less than 30 seconds is 0.75. Assume that your calls are independent.
 - a. If you call 10 times, what is the probability that exactly 9 of your calls are answered within 30 seconds?
 - b. If you call 20 times, what is the probability that at least 16 calls are answered in less than 30 seconds?
 - c. If you call 20 times, what is the mean number of calls that are answered in less than 30 seconds?
6. The number of failures of a testing instrument from contamination particles on the product is a Poisson random variable with a mean of 0.05 failures per hour.
 - a. What is the probability that the instrument does not fail inane 12-hour shift?
 - b. What is the probability of at least one failure in a 20-hourday?

Practice questions Normal distribution

1. The length of an injection-molded plastic case that holds magnetic tape is normally distributed with a length of 90.2 millimeters and a standard deviation of 0.1 millimeter.
 - (a) What is the probability that a part is longer than 90.3 millimeters or shorter than 89.7 millimeters?
 - (b) What should the process mean be set at to obtain the greatest number of parts between 89.7 and 90.3 millimeters?
2. The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.
 - (a) What is the probability that a laser fails before 500 hours?
 - (b) What is the life in hours that 95% of the lasers exceed?
3. A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live
 - (a) more than 32 months;
 - (b) less than 28 months; (c) between 37 and 49 months.
4. The average grade for an exam is 50, and the standard deviation is 5. If 14% of the class is given A_s and the grades are curved to follow a normal distribution, what is the lowest possible A ?

Practice questions Central limit theorem

1. The elasticity of a polymer is affected by the concentration of a reactant. When low concentration is used, the true mean elasticity is 55, and when high concentration is used the mean elasticity is 60. The standard deviation of elasticity is 4, regardless of concentration. If two random samples of size 16 are taken, find the probability that $\bar{X}_{high} - \bar{X}_{low} \geq 2$
2. A consumer electronics company is comparing the brightness of two different types of picture tubes for use in its television sets. Tube type A has mean brightness of 100 and standard deviation of 16, while tube type B has unknown mean brightness, but the standard deviation is assumed to be identical to that for type A. A random sample of $n = 25$ tubes of each type is selected, and $\bar{X}_B - \bar{X}_A$ is computed. If μ_B equals or exceeds, μ_A the manufacturer would like to adopt type B for use. The observed difference is $\bar{X}_B - \bar{X}_A = 3.5$ what decision would you make, and why?
3. Two distinct solid fuel propellants, type A and type B are being considered for a space program activity. Burning rates of the propellant are crucial. Random samples of 20 specimens of the two propellants are taken with Population means 20.5 cm/sec for propellant A and 24.50 cm/sec for propellant B. It is generally assumed that the variability in burning rate is roughly the same for the two propellants and is given by a population standard deviation of 5 cm/sec. Assume that the burning rates for each propellant are approximately normal

$$P(\bar{x}_B - \bar{x}_A \geq 4 \cdot 9)$$

4. A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with mean 70 and standard deviation 12. let $\bar{X}_1$ and $\bar{X}_2$ be the two sample mean find the probability that
 - a. The probability that $\bar{X}_1 - \bar{X}_2$ exceed 4.
 - b. The probability that $3.5 \leq \bar{X}_1 - \bar{X}_2 \leq 5.5$
5. Two independent experiments are run in which two different types of paint are compared. Eighteen specimens are painted using type A, and the drying time, in hours, is recorded for each. The same is done with type B. The population standard deviations are both known to be 1.0. Assuming that the mean drying time is equal for the two types of paint, find

$$P(\bar{x}_A - \bar{x}_B \geq 1 \cdot 0)$$

Where $n_1 = n_2 = 18$

Practice questions of Byes theorem

1. Because a new medical procedure has been shown to be effective in the early detection of an illness, a medical screening of the population is proposed. The probability that the test correctly identifies someone with the illness as positive is 0.99, and the probability that the test correctly identifies someone without the illness as negative is 0.95. The incidence of the illness in the general population is 0.0001. You take the test, and the result is positive. What is the probability that you have the illness?
2. An inspector working for a manufacturing company has a 99% chance of correctly identifying defective items and a 0.5% chance of incorrectly classifying a good item as defective. The company has evidence that its line produces 0.9% of Nonconforming items.
 - (a) What is the probability that an item selected for inspection is classified as defective?
 - (b) If an item selected at random is classified as non defective, what is the probability that it is indeed good?
3. In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer?
4. Semiconductor lasers used in optical storage products require higher power levels for write operations than for read operations. High-power-level operations lower the useful life of the laser. Lasers in products used for backup of higher speed magnetic disks primarily write, and the probability that the useful life exceeds five years is 0.95. Lasers that are in products that are used for main storage spend approximately an equal amount of time reading and writing, and the probability that the useful life exceeds five years is 0.995. Now, 25% of the products from a manufacturer are used for backup and 75% of the products are used for main storage.
 - a. What is the probability that the useful life of a laser exceeds five years?
 - b. What is the probability that a laser that failed before five years came from a product used for backup?